SET A

AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH
Faculty of Science and Technology
Department of Computer Science
CSC 1205 Object Oriented Programming - 1 (Section: All)

Mid Term Examination **Summer 2022-23**

Total Marks: 100 **Moderator:** RIFATH MAHMUD **Time:** 2 hours

General Instructions:

1. All the questions are based on OBE program outcomes. **Question number 31 in Part B is mandatory and is mapped to CO number 1 (You must rewrite you Name, ID, Section, and Date on Part B).**
2. Use pencil/pen to write the answer. Return the question paper, OMR sheet at the end of the examination.
3. Use ball point pen/Pencil to fill the OMR sheet. Do not staple/bend/fold the OMR sheet.

Specific Instructions:

1. Answer Part-A in the OMR. Answer Part-B, Part-C, Part-D & Part-E in the Question Paper within the given space.
2. **Do not staple OMR with this question paper.**
3. You may take signed loose sheet(s) from the Invigilators to answer Part-E. Staple only the loose sheet with this question paper that you have used for Part-E(code writing).

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Part – A (Answer All) **MCQ** **[1 x 30 = 30]**

Use the following two options for questions 1 to 10.
a) True b) False

1. Objects of a particular class maintain different memory locations for the instance variables. T
2. Encapsulation is achieved by declaring all the data members as public. F
3. Procedural programming provides modularity and data security. T
4. In Java, the main function is static because no objects are required to invoke it. T
5. Inheritance provides code reusability. T
6. In Java, default access modifier can be applied at class and member level T
7. Static variables must be declared as public. F
8. More than one public class is allowed in a single java file. F
9. In Java, objects of String class are mutable. T /F
10. Multiple static blocks execute sequentially based on the declaration. T
11. A Java Byte Code can be run on any platform having a compatible version of ________
    a) JDK b) JRE c) JAVA Compiler d) Option a and b
12. Which one of the following is not a primitive data type?
    a) int b) double c) String d) bool
13. In Java, which option is valid for getting the size of a String?
    a) length b) Size c) length() d) count
14. The String `compareTo()` method returns ____?
    a) true, false b) -1, 0, 1 c) Less than 0, 0, greater than 0 d) null
15. Which one will check whether str1 and str2 contain the same value or not?
    a) str1.equal(str2) b) str1= str2 c) str1.equals(str2) d) str1== str2

16. The follow code segments outputs ____________.

```
String s1 = "OOP1"; String s2 = "oop2";
System.out.println(s1.compareTo(s2));
```

a) 0 b) a value greater than 0 c) a value less than 0 d) false

17. What is the correct output of the following code segment?

```
int i = (float)225;
float f = i;
System.out.println(f);
```

a) 225 b) 225.0 c) 225.522 d) Compilation Error

18. Which of the following is an explicit call to the superclass's default constructor?

a) `default();` b) `class();` c) `super();` d) `base();`

19. _______ access modifier allows accessibility within all classes in the same package and within subclasses in other packages.

a) private b) protected c) public d) package-private

20. Car is a type of Vehicle. Which OOP principle is indicated here?

a) encapsulation b) inheritance c) polymorphism d) abstraction

21. Which variation of inheritance in not supported by java.

a) single b) multi-level c) hierarchical d) multiple

22. Identify the correct way of declaring constructor for a class named `Entity`

a) `Entity(){}` b) public `Entity(){}` c) void `Entity(){}` d) Both a and b

23. Hide the last 5 digits of the phone number as *+88017123******

Choose the appropriate parameters for the substring method to generate the output.

```
String phoneNumber = "+8801712345678";
System.out.println(phoneNumber.substring(0 , 9 )+"*****");
```

a) 1,9 b) 0,8 c) 0,9 d) None

24. An array can store more than one type of data.

a) Valid b) Invalid

25. Which of these is a legal declaration an array?

a) `int []arr = new int[]{-1,-2};` b) `float []arr = (7,8);`
c) `char arr[] = [4, 5];` d) `double arr[] = new double{2.5, 8.5};`

26. Static block is usually used to initialize__________.

a) Instance variables b) class variables c) objects d) default values

27. Which of these has the responsibility of providing controlled access to the data of a class?

a) Parent class b) Attributes c) Methods d) this keyword

28. What is the correct import statement to import Scanner class in java?

a) import java.lang.Scanner; b) import java.util.Scanner; c) import util.Scanner d) None

29. Considering sc is an object of the Scanner class, which will be more appropriate for the following line?

String message = sc.__________________();

a) `nextInt();` b) `nextDouble()` c) `nextLine();` d) `nextAnything();`

30. "Officer is an Employee". Here___

a) Officer is the super class of Employee b) Employee is the base class of Officer
c) Employee is the derived class of Officer d) Both Officer and Employee are child classes

Ans to the McQ

1. T
2. F
3.F
4.T
5.T
6.T
7. F
8.F
9. F
10. T
11. a
12. c
13. c
14.c
15. c
16.c
17. d
18. c
19. b
20. b
21. d
22. D
23. C
24. B
25. A
26. B
27. C
28. B
29. C
30. B

Part – C (Answer 1 out of the 2) **Short Question** **[1 x 5 = 5]**

32. ***Complete the following code sample to ask the user what is the name and age of a person? Then store and display the details.*** ***(5)***

```
import java.util.Scanner;
public class Person {
    private String name;
    private int age;

    public Person(String name, int age) {
        this.name = name;
        this.age = age;
    }
    public void displayInfo() {
        System.out.println("Name: " + name);
        System.out.println("Age: " + age);
    }
    public static void main(String[] args) {
        Scanner scanner = new Scanner (System.in);
        System.out.print("Enter name: ");
        String name = scanner.nextLine();
        System.out.print("Enter age: ");
        int age = scanner.nextInt();

        Person person = new Person(name, age);
        person.displayInfo();
    }
}
```

5

33. **Identify the type of type casting as implicit or explicit and provide examples for the following scenario.** **(5)**

I. byte to short
Type of casting: **Example:**

ii. float to double
Type of casting: **Example:**

Q. Consider the following class descriptions – ***Teacher*** and ***Student***. Explain clearly what the real problem is in the context of Object Oriented Programming Principles. Provide a solution that incorporates appropriate Object Oriented Programming Principle(s).

Teacher
`String name;`
`int age;`
`double salary;`
`void setName(String name)`
`void setAge(int age)`
`void setSalary(double salary)`

Student
`String name;`
`int age;`
`double CGPA;`
`void setName(String name)`
`void setAge(int age)`
`void setCGPA(double CGPA)`

A. Problem Identification in the context of Object-Oriented Principle(s). Write your answer below.

➔ The problem in the provided class descriptions is a lack of encapsulation and Inheritance. In object-oriented programming, encapsulation, and Inheritance are essential principles that help in creating well-structured and maintainable code. Let's identify the specific issues:

1. Lack of Encapsulation: In both the Teacher and Student classes, the instance variables (name, age, salary for Teacher, and name, age, CGPA for Student) are declared as public, which violates the principle of encapsulation. Encapsulation promotes the idea that the internal state of an object should be hidden from external access, and it should be accessed through methods to control and validate the data.

2. Duplication of Code: Both classes have similar attributes (name and age) and similar setter methods (setName and setAge). This duplication of code violates the DRY (Don't Repeat Yourself) principle, a core principle of software development.

3. Lack of Inheritance: There is no indication of any inheritance or relationship between the Teacher and Student classes. Object-oriented programming encourages the use of inheritance to represent relationships between classes and to promote code reusability.

B. Provide a solution that incorporates appropriate Object-Oriented Programming Principle(s). Write your answer below.

➔ To address the identified problems, we can apply appropriate object-oriented programming principles:

1. Encapsulation: Make the instance variables private and provide public methods (getters and setters) to access and modify them. This ensures that the internal state of the objects is protected and controlled.

2. Inheritance: Introduce a common base class **Person** that encapsulates common properties (name and age) and then create specialized classes (Teacher and Student) that inherit from the **Person** class. This promotes code reusability and represents the "is-a" relationship between Teacher/Student and Person.

```
public class Person {
  private String name;
  private int age;

  public String getName() {
    return name;
  }

  public void setName(String name) {
    this.name = name;
  }
```

```
    public int getAge() {
        return age;
    }

    public void setAge(int age) {
        this.age = age;
    }
}

public class Teacher extends Person {
    private double salary;

    public double getSalary() {
        return salary;
    }

    public void setSalary(double salary) {
        this.salary = salary;
    }
}

public class Student extends Person {
    private double CGPA;

    public double getCGPA() {
        return CGPA;
    }

    public void setCGPA(double CGPA) {
        this.CGPA = CGPA;
    }
}
```

With these changes, we have improved encapsulation, reduced code duplication, introduced inheritance, and adhering to object-oriented programming principles.

Part - D (Answer 2 out of 3) **Output Tracing** [2 x 15 = 30]

34.
```
class ClassA{
 static{
 System.out.print("Loading Class: A ");
 }

 public ClassA(){
 System.out.println("Constructor: A");
 }
}

class ClassB{
 static{
 System.out.print("Loading Class: B ");
 }

 public ClassB(){
 System.out.println("Constructor: B");
 }
}

public class Main{
 public static void main(String []args){
  ClassB cB = new ClassB();
  cB = new ClassB();
  ClassA cA = new ClassA();
 }
 static{
 System.out.println("Loading Class: Main");
 }
}
```

:B

Loading class main
Loading class B
Constractor: B
Constractor: B
Loading class A
Constractor: A

35.
```
public class Main {
 public static void main(String args[ ]){
  int array[] = {10,27,2,15,7,9,30,23,12,21};
  float count = 0;
  float[ ] newarray = new float[10];

  for(int i=0;i<array.length;i++) {
      if(array[i]%3 == 0) {
              count++;
              newarray[i]=array[i];
      }
  }
  System.out.println("Count:"+ count);
  System.out.print("Elements:");

  for(int j = 0;j<newarray.length;j++) {
              System.out.println(newarray[j]);
  }
 }
}
```

35. Outputs

Count: 6.0
Element: 0.0
27.0
0.0
15.0
0.0
9.0
30.0
0.0
12.0
21.0

36.
```
import java.lang.*;
public class Strings {
public static void main(String args[ ]) {
    String str1="OBJECT";
    String str2= new String("java,util,Scanner");
    String str3= str2.substring(4,10);   16
    String str4="OBJECT ORIENTED";   .util
    String str5 = new String("PROGRAMMING");

    System.out.println("Length: "+str3.length());
    System.out.println(str1.compareTo(str4));
    System.out.println(str2.indexOf(','));
    System.out.print(str5.charAt(9));


    String str6=str1.toUpperCase(); OBJECT

    if(str1.equals(str6)){
        System.out.println("false"); false
    }
    else{
        System.out.println("true");
    }
  }
}
```

36. Outputs

Length:6
-9
4
Nfalse

Part - E Code Writing **[1 x 25 = 25]**

MobilePhone
String brand; String model; double batteryCapacity;
MobilePhone() **MobilePhone**(String brand,String model, double batteryCapacity)
void setBrand(String brand) String getBrand() void setModel(String model) String getModel() void setBatteryCapacity(double batteryCapacity) double getBatteryCapacity() void showDetails()

SmartPhone
int cameraResolution; int RAMSize; int ROMSize; String OSVersion;
SmartPhone() **SmartPhone**(String brand,String model, double batteryCapacity, int cameraResolution, int RAMSize, int ROMSize,String OSVersion)
void setCameraResolution(int cameraResolution) int getCameraResolution() void setRAMSize(int RAMSize) int getRAMSize() void setROMSize(int ROMSize) int getROMSize() void setOSVersion(String OSVersion) String getOSVersion() void showDetails()

Note: You should complete the **MobilePhone and SmartPhone class** and after that write a Main Class and create **two Sample objects** of the **SmartPhone** class and demonstrate the use of different methods. ***(Start writing your code on loose sheet...)***